with respect to the naïve version, we will highlight those changes in red, to help readers quickly compare the two versions.

Instead of mapping to an actual set, elements in the partition map points to each element's parent in the tree. That's why, as you can see in the book's repo, we can rename our `partitionsMap` to `parentsMap`, to make its purpose explicit.

At initialization, we conveniently set each element as its own parent: trust me on that one, we'll see why later.

The same change applies to the `add` method, which stays otherwise unchanged.

The `findPartition` method, instead, needs quite some tuning to properly work. Two notes on its implementation:

With respect to the basic implementation, in this case we won't return a list anymore, but rather the element at the root of the partition's tree. The return value of `findPartition` might not immediately make sense to an external caller, and in fact this method will mostly be used internally, when called by merge and `areDisjoint` methods.

Listing 5.6 Tree-based solution, findPartition

```
class DisjointSet
  #type HashMap[Element, Tree[Element]]
  parentsMap

function
findPartition(elem)
      #1
```